

Basics of Web Hosting and DNS

Resource Person: Server Administrator, IPSR Solutions Limited

1. What is Web Hosting

Web hosting is a service that stores a website's files on a server and makes them accessible over the internet via a domain name.

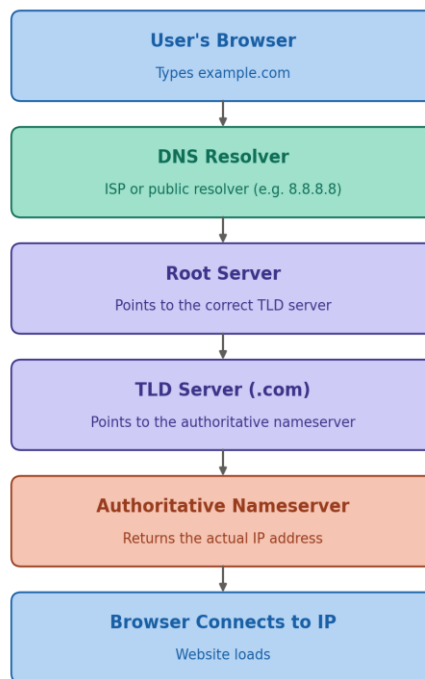
Type	Description
Shared hosting	Multiple websites share one server's resources — cheapest, least control
VPS hosting	A virtual slice of a server dedicated to you — more control, more cost
Dedicated hosting	An entire physical server just for you — full control, highest cost
Cloud hosting	Resources spread across multiple servers (e.g. AWS EC2) — scalable, pay-as-you-go

2. What is DNS (Domain Name System)

DNS translates human-readable domain names, such as example.com, into the IP addresses that computers use to locate servers.

DNS Resolution Flow

The diagram below shows the path a domain lookup takes, from the user's browser through the resolver, root and TLD servers, down to the authoritative nameserver that holds the actual IP address, and finally back to load the website.



Result is cached along the way for faster future lookups

Step-by-step explanation

- The user's browser sends a request to resolve the typed domain name.
- A DNS resolver, usually run by the ISP or a public service, receives the request.
- The resolver queries a root server, which points to the correct TLD server.
- The TLD server (for example .com) points to the domain's authoritative nameserver.
- The authoritative nameserver returns the actual IP address for the domain.
- The browser connects directly to that IP address, and the website loads.
- The result is cached at multiple levels so future lookups are faster.

3. Key DNS Record Types

Record	Purpose
A	Maps a domain to an IPv4 address
AAAA	Maps a domain to an IPv6 address
CNAME	Points one domain or subdomain to another domain name
MX	Directs email for the domain to a mail server
TXT	Stores text data, often used for SPF, DKIM, or domain verification
NS	Specifies the authoritative nameservers for the domain
PTR	Reverse lookup — maps an IP back to a domain, used for mail server verification
SOA	Start of Authority — holds admin info about the domain's DNS zone (primary nameserver, admin email, refresh timers)
SRV	Points to a specific service and port, e.g. for VoIP or chat applications
CAA	Restricts which certificate authorities are allowed to issue SSL/TLS certificates for the domain

4. Domain Registration Flow

- Choose a domain name and check availability via a registrar (GoDaddy, Namecheap, Route 53, etc.).
- Purchase and register the domain.
- Point the domain's nameservers to the hosting provider or DNS provider.
- Add DNS records: an A record to the server's IP, MX records for email, and so on.
- Wait for DNS propagation — changes can take from a few minutes up to 48 hours to update worldwide, since records are cached at multiple levels.

5. Connecting Hosting and DNS in Practice

- Spin up a server, for example an AWS EC2 instance.
- Note its public IP address.
- In the domain registrar's DNS settings, create an A record pointing the domain to that IP.
- Install a web server, such as Apache or Nginx, on the instance.
- Once DNS propagates, visiting the domain loads the server's content directly.

6. Common DNS Troubleshooting Commands

Command	Purpose
nslookup example.com	Check what IP a domain resolves to
dig example.com	More detailed DNS lookup

dig example.com MX	Check mail records specifically
whois example.com	Check domain registration information
ping example.com	Confirm the domain resolves and the server responds